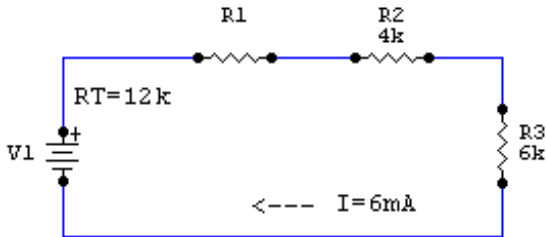


Week 1 Problems -- SOLUTIONS

1. Ohm's Law:



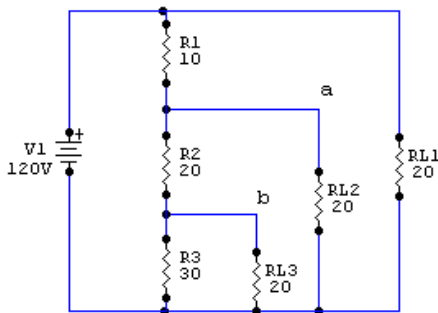
For the circuit shown, on the left, determine the values of the voltage V_1 and the resistor R_1

$$R_T = R_1 + R_2 + R_3 \rightarrow 12 \text{ k}\Omega = R_1 + 4 \text{ k}\Omega + 6 \text{ k}\Omega$$

$$\text{So, } \underline{R_1 = 2 \text{ k}\Omega}$$

$$\underline{V_1 = I \times R_T = 6 \text{ mA} \times 12 \text{ k}\Omega = 72 \text{ V}}$$

2. Voltage Divider Rule:



For the circuit shown on the left, determine the values of the voltages at points 'a' and 'b'. Determine also the current drawn from the battery.

$$\text{At pt. b, } R_{L3} \parallel R_3 \rightarrow 20 \Omega \parallel 30 \Omega = 12 \text{ k}\Omega$$

$$\text{At pt. a, } (R_2 + 12) \parallel R_{L2} \rightarrow 32 \Omega \parallel 20 \Omega = 12.31 \Omega$$

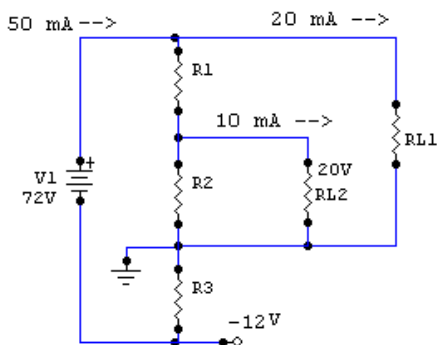
Using the voltage divider rule,

$$\underline{V_a = 120 \text{ V} \times 12.31 / (12.31 + 10) = 66.21 \text{ V} \leftarrow}$$

$$\underline{V_b = 66.21 \text{ V} \times 12 / (12 + 20) = 24.83 \text{ V} \leftarrow}$$

$$\underline{I_{\text{Batt}} = V_1 / R_{L1} + V_1 / (10 + 12.31) = 11.38 \text{ A} \leftarrow}$$

3. Kirchhoff's Current Law:



For the circuit shown on the left, determine the values of the resistors R_1 , R_2 and R_3 .

Can 2W resistors be used in the design?

$V_{RL1} = 72 - 12 \text{ V} = 60 \text{ V}$, since the 'ground' is at the lower node of (R_2 , $RL2$, $RL1$).

From K.C.L., $I_{R1} = 50 \text{ mA} - 20 \text{ mA} = 30 \text{ mA}$, and

$V_1 = (60 - 20) \text{ V} = 40 \text{ V}$, since there are 20 V across $RL2$.

$$\text{So, } \underline{R_1 = 40 \text{ V} / 30 \text{ mA} = 1.33 \text{ k}\Omega \leftarrow}$$

Again, K.C.L., $I_{R2} = 30 \text{ mA} - 10 \text{ mA} = 20 \text{ mA}$, and $V_{R2} = 20 \text{ V}$

$$\text{So, } \underline{R_2 = 20 \text{ V} / 20 \text{ mA} = 1 \text{ k}\Omega \leftarrow}$$

$I_{R3} = 50 \text{ mA}$, and $V_{R3} = 12 \text{ V}$

$$\text{So, } \underline{R_3 = 12 \text{ V} / 50 \text{ mA} = 240 \Omega \leftarrow}$$

$RL1 = 60 \text{ V} / 20 \text{ mA} = 3 \text{ k}\Omega$ and $RL2 = 20 \text{ V} / 20 \text{ mA} = 1 \text{ k}\Omega$

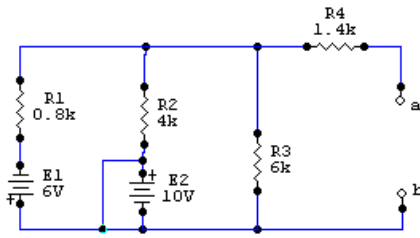
Power dissipation: $(I^2 \times R) \text{ Watts}$

$R_1 (1.33 \text{ k}\Omega, 30 \text{ mA}) \rightarrow 1.197 \text{ W}$; $R_2 (1 \text{ k}\Omega, 20 \text{ mA}) \rightarrow 0.4 \text{ W}$; $R_3 (240 \Omega, 50 \text{ mA}) \rightarrow 0.6 \text{ W}$;

$RL1 (3 \text{ k}\Omega, 20 \text{ mA}) \rightarrow 1.2 \text{ W}$; $RL2 (2 \text{ k}\Omega, 10 \text{ mA}) \rightarrow 0.2 \text{ W}$

Therefore, 2 W resistors may be used, since the maximum power dissipated is 1.2 W in $RL1$.

4. Thévenin, Superposition:



Find the Thévenin equivalent circuit between points 'a', 'b'.

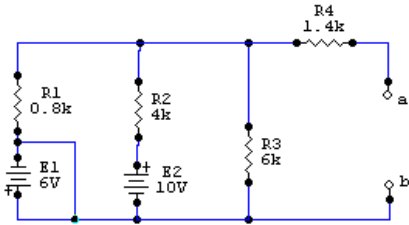
For the Thévenin voltage:

Use Superposition, considering first E1 only:

With E2 shorted, R2 & R3 are in parallel $\rightarrow 4k\Omega \parallel 6k\Omega = 2.4k\Omega$.

R4 is 'hanging' and can be ignored.

Therefore, $E_{th1} = -6V \times (2.4 / (2.4+0.8)) = -4.5 V$



With E1 shorted, R1 & R3 are in parallel $\rightarrow 0.8k\Omega \parallel 6k\Omega = 0.706k\Omega$.

Therefore, $E_{th2} = 10V \times (0.706 / (0.706+4)) = +1.5 V$

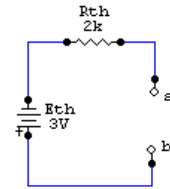
The sum of the voltages is $E_{th} = -3 V$

For the Thévenin resistance, short both E1 & E2.

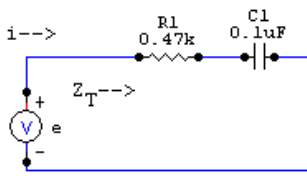
It is seen that R1, R2 & R3 are in parallel $\rightarrow$ result = 0.6 kΩ

Therefore, $R_{th} = 1.4k\Omega + 0.6k\Omega = 2 k\Omega$

The Thévenin equivalent circuit is shown on the right.



5.



$e = 20\sin(\omega t) V$, and $f = 1 kHz$

[Note that $\omega = 2\pi f$ rad/sec]

(a) Determine Z_T (b) Find the circuit current, i .

(b) Calculate V_R and V_C (d) Draw the phasor diagram for e , V_R and V_C

$$Z_T = R1 + (1/(j \omega C)) = 470 - j1591.6 \rightarrow 1659.5 \Omega / -73.55^\circ$$

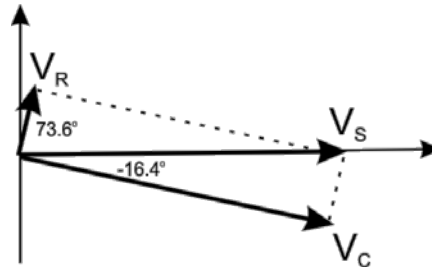
$$e_{rms} = 20 \times 0.707 = 14.14 V \text{ (RMS)}$$

$$i = 14.14V / (1659.5 \Omega / -73.55^\circ) = 8.52 \text{ mA} / +73.55^\circ$$

$$V_R = I \times R1 = 8.52 \text{ mA} / +73.55^\circ \times 470 = 4.004 V / +73.55^\circ$$

$$V_C = I \times Z_C = 8.52 \text{ mA} / +73.55^\circ \times (-j1591.6) = 13.56 V / -16.44^\circ, \text{ since } -j = 1 / -90^\circ$$

The phasor diagram:



6. $R = 10 \text{ k}\Omega$, $C = 0.1 \text{ }\mu\text{F}$, $V_b = 5 \text{ V}$.

The time constant, $\tau = (10 \times 10^3) \times (0.1 \times 10^{-6}) = 10^{-3} = 1 \text{ ms}$

$$\text{When } t = 0.5\tau, \quad V_c = V_b \left(1 - e^{\frac{-0.5\tau}{\tau}} \right) = 5(1 - e^{-0.5}) = \mathbf{1.967 \text{ V}}$$

$$\text{Switch in Pos. 2, at } t = 0.2 \text{ ms: } -i = \frac{V_b}{R} e^{\frac{-0.2}{1.0}} = \frac{5V}{10k\Omega} e^{-0.2} = \mathbf{0.409 \text{ mA}}$$

- 7.

$$\text{The expression } \frac{(6 + j5)(10 - j6)}{(4 - j3)(5 + j5)} = 2.576 + j0.032$$